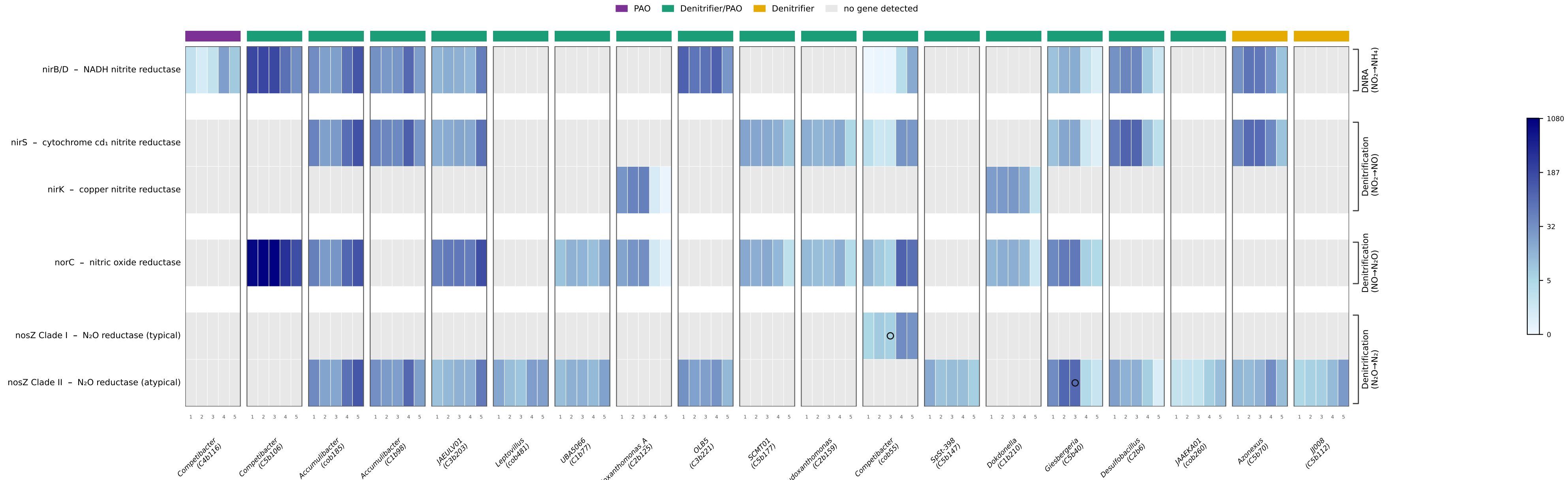


Denitrification & DNRA gene abundance across top high-quality MAGs of the CAN system



Per-sample gene abundances in reads per kilobase per million mapped reads (RPKM) from long-read (Nanopore) metagenomes of the CAN system. Columns are grouped by MAG — the union of the ten most abundant high-quality MAGs per sample — each subdivided into the five samples CAN_1-CAN_5 (sub-columns 1-5, left→right); the coloured bar marks GAO / PAO / denitrifier classification (mag_abundance_summary.tsv logic). Rows are denitrification/DNRA KO functions grouped by nitrogen-transformation stage (right brackets); for nirB/D (K00362/K00363) the cell is the maximum RPKM over the two genes. nosZ (K00376) is split into Clade I (typical denitrifier) and Clade II (atypical / non-denitrifier) by per-gene nhmmer classification against the NosZREF Clade I and Clade II (sub-clades 1577 A-H) reference alignments; ○ marks a low-confidence clade call. Grey cells indicate no detected gene.